



FLTA25

FEDERATED DISTILLATION FOR EDGE DEVICES: EFFICIENT CLIENT-SIDE FILTERING FOR NON-IID DATA

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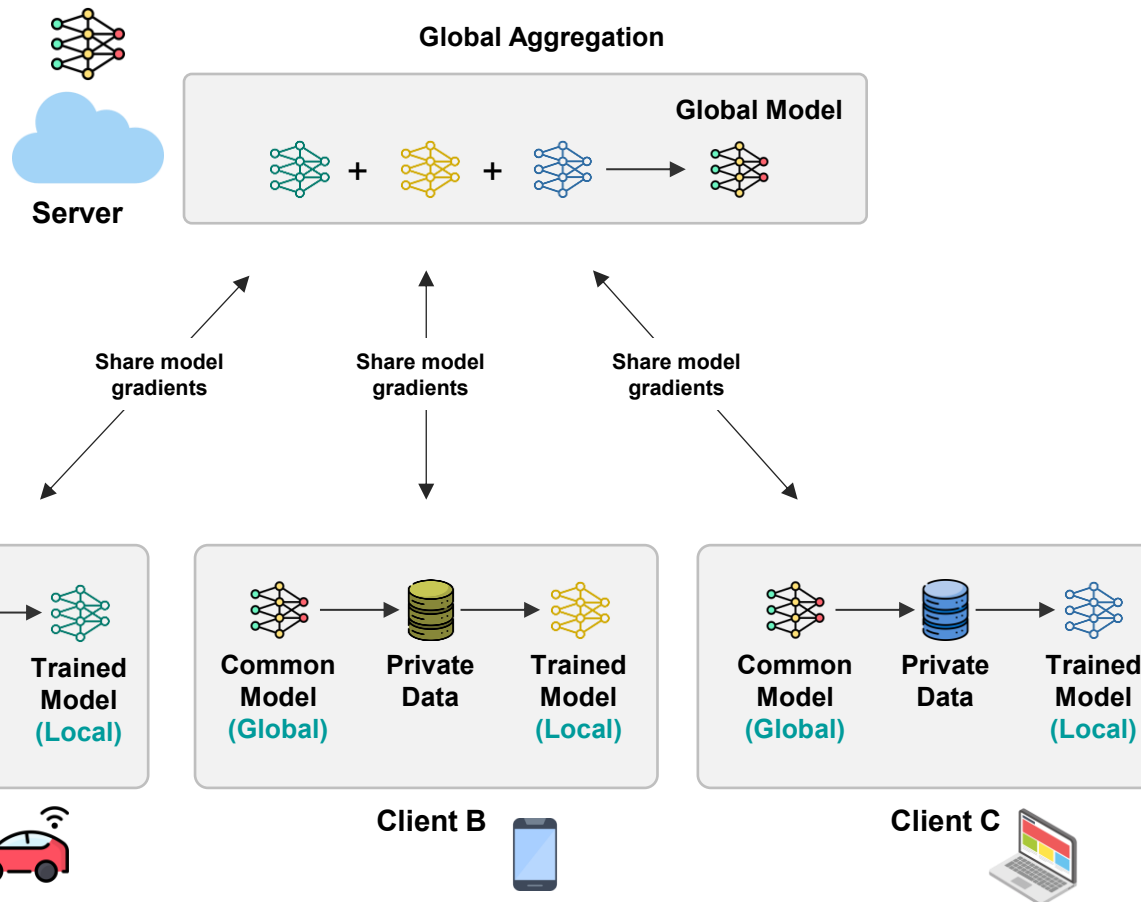
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FEDERATED LEARNING

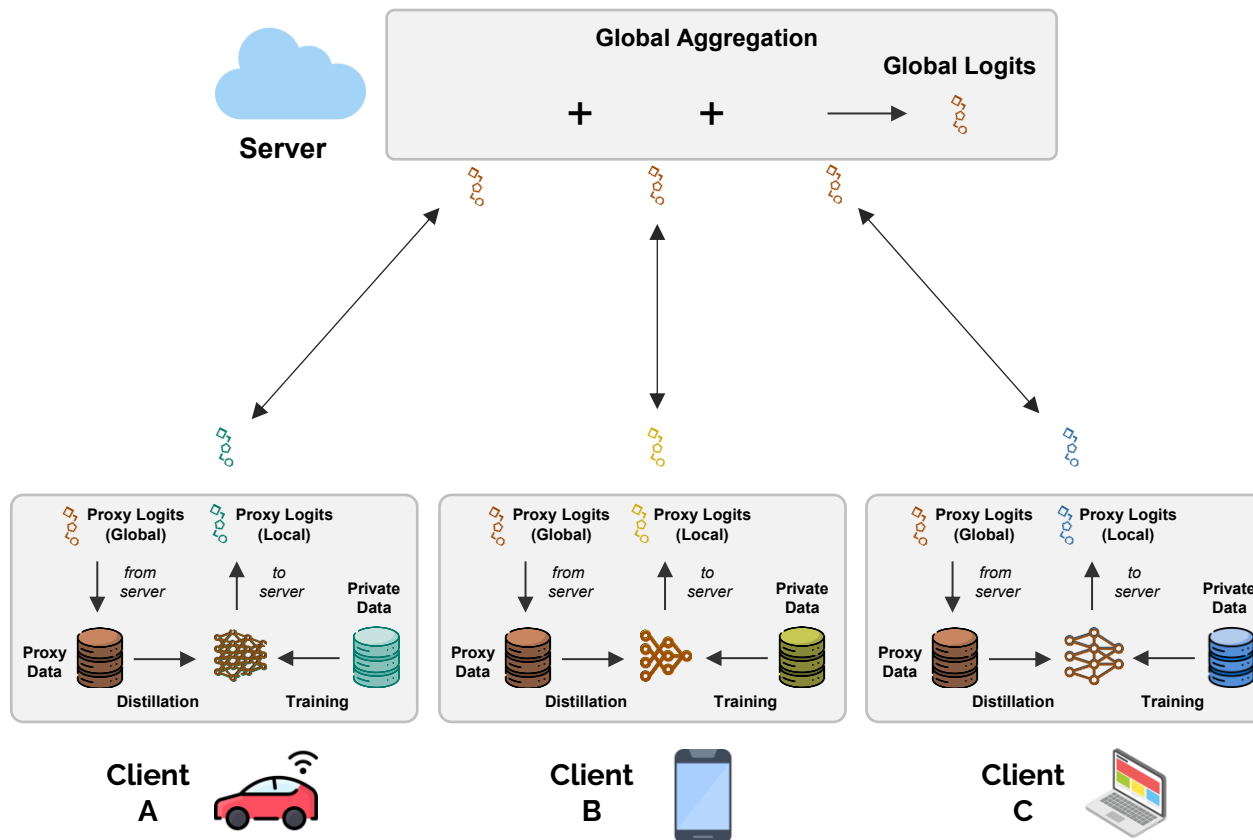


LEARN COLLABORATIVELY BY SHARING MODEL GRADIENTS WITHOUT SHARING USER DATA [1].

Issues

- × **Model privacy** (private information can still be retrieved from shared model gradients).
- × No **model heterogeneity** (shared common model among clients).
- × **Communication costs** (infeasible for resource-constrained edge devices).
- × Sub-optimal performance on **non-IID** data.

FEDERATED DISTILLATION



AN AMALGAMATION OF FEDERATED LEARNING AND KNOWLEDGE DISTILLATION THAT ENABLES COLLABORATIVE LEARNING BY SHARING MODEL LOGITS ONLY [1].

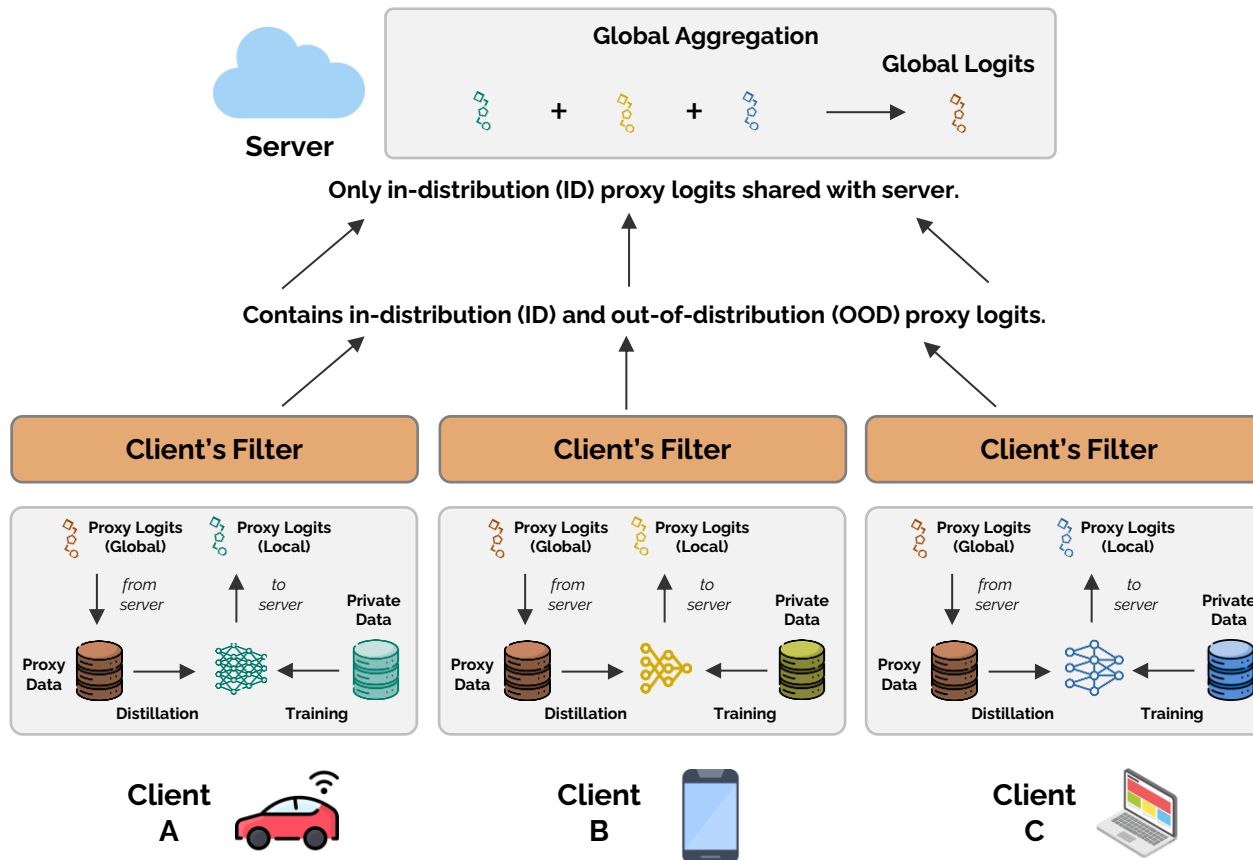
Advantages:

- ✓ **Model heterogeneity**: Each client has customized model.
- ✓ **Low communication cost** (only logits or intermediate features).

Disadvantages

- ✗ Sub-optimal performance on **non-IID** data.
- ✗ **Increases the load** on edge devices (training + inference + knowledge distillation).

SELECTIVE FEDERATED DISTILLATION

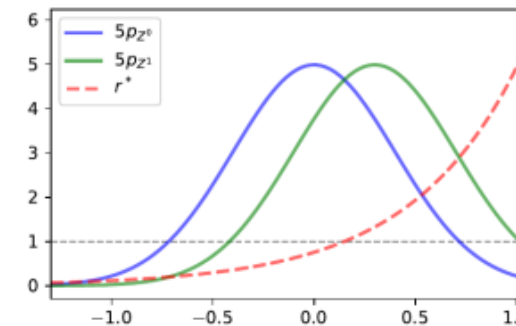


ID: Proxy samples that lies in private data distribution.
OOD: Proxy samples that DO NOT lies in private data distribution.

Problem: Client models typically generate **wrong predictions on OOD proxy samples**, resulting in sub-optimal performance.

SHARE **ID PROXY LOGITS** WITH SERVER TO ACHIEVE BETTER PERFORMANCE ON NON-IID DATA [1].

→ **Filter out-of-distribution proxy logits.**



Density Ratio Estimator [2]

$$r^* := \frac{p_o(z)}{p_1(z)}$$

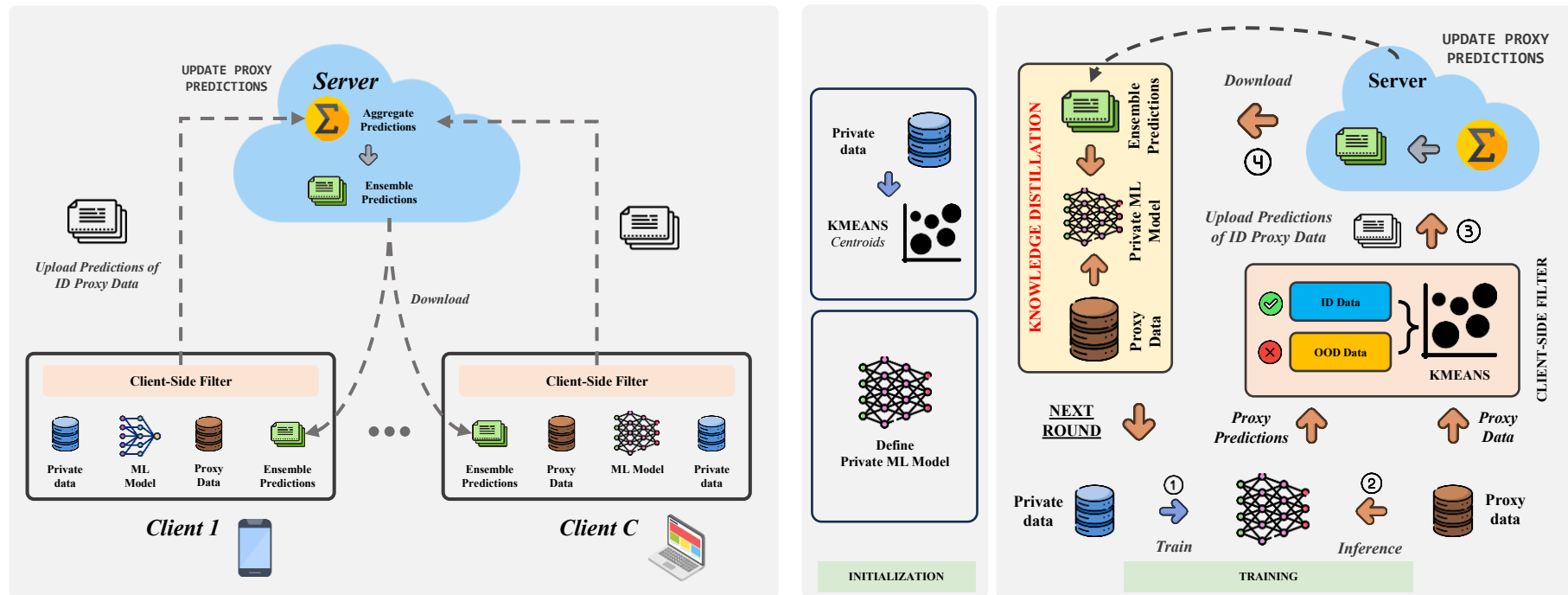


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EDGE-FD: EFFICIENT CLIENT-SIDE FILTERING FOR NON-IID DATA

PROPOSED SOLUTION

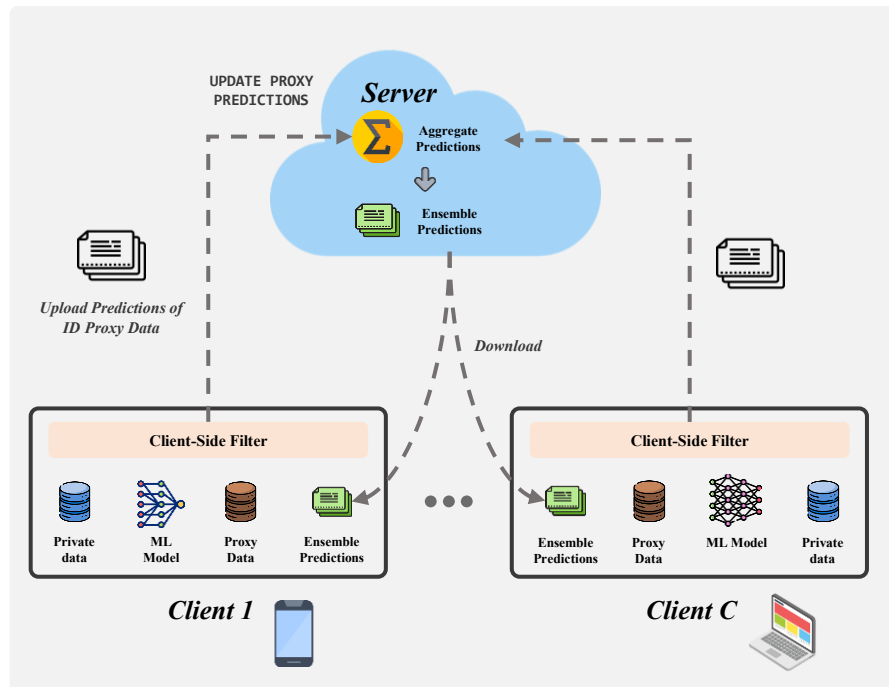
EDGE-ORIENTED FEDERATED DISTILLATION



EdgeFD: A federated distillation framework using KMeans density ratio estimation to effectively filter OOD proxy logits on heterogeneous devices, customized ML models, and personalized user data.

EDGE-ORIENTED FEDERATED DISTILLATION

Global View



Client: All clients contains the following:

- Private data
- Customized ML model
- Proxy data
- Ensemble predictions (Global logits)
- Client-side filter

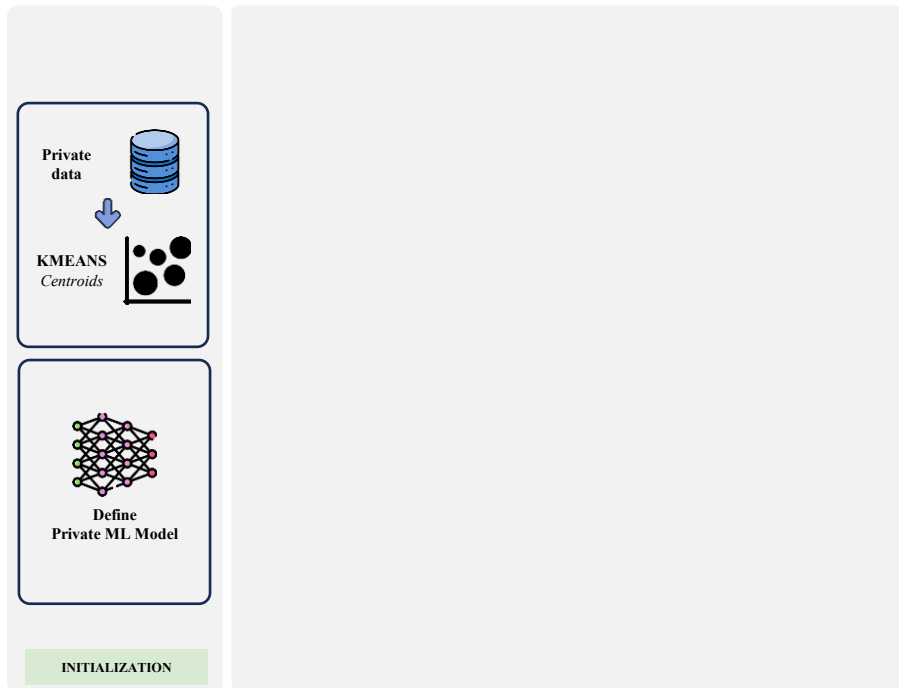
All clients upload in-distribution predictions of proxy data to server.

Server:

- Server aggregates the in-distribution predictions.
- All clients download the ensembled proxy predictions (global logits) from the server for knowledge distillation.

EDGE-ORIENTED FEDERATED DISTILLATION

Client's View



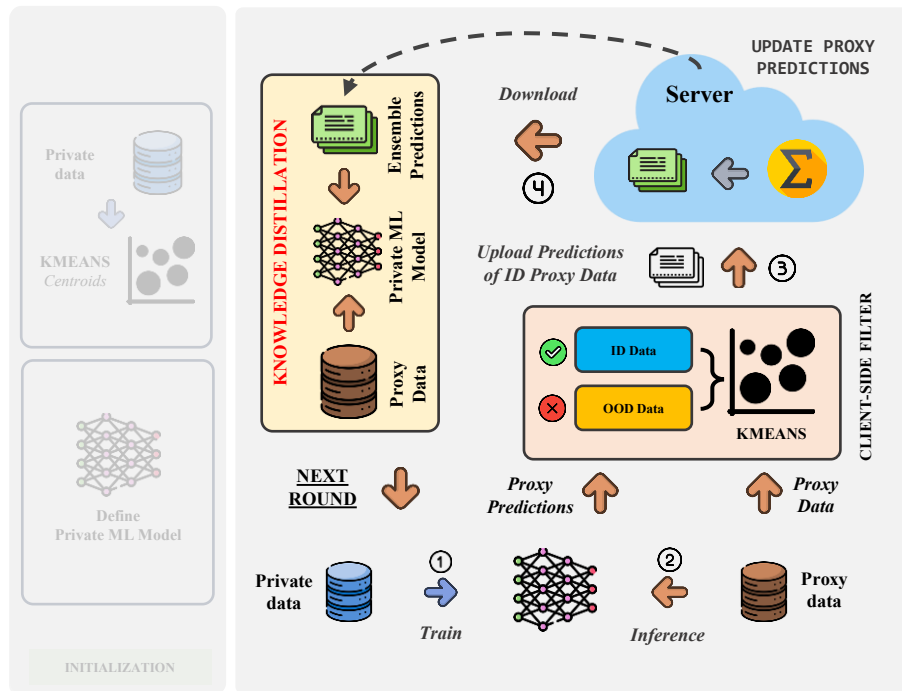
Initialization phase

All clients initialize the following:

- Customized ML model w.r.t the available resources.
- Trains the KMeans model on private data for density ratio estimation.
- **Proxy Data:** Share a percentage of private data for the generation of proxy data on the server.
- Download the proxy data from the server.

EDGE-ORIENTED FEDERATED DISTILLATION

Client's View



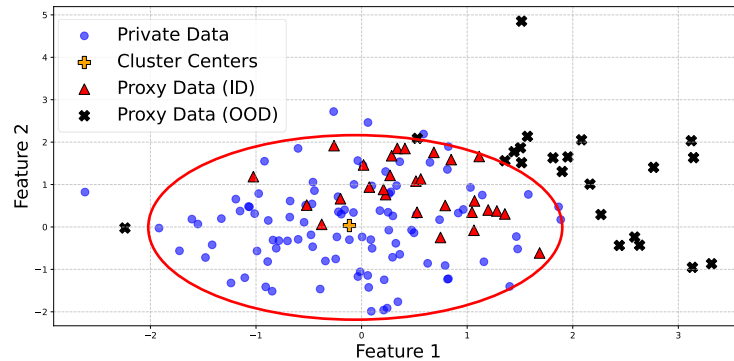
Training Phase

All client(s) perform the following:

1. Train private ML model on private data.
2. Inference on proxy data and submits them to the filter.
3. The client-side filters classifies ID and OOD proxy samples as follows:
 - Calculates the Euclidean distance between proxy sample and learned centroid position on private data.
 - Two staged Filter: Proxy sample is classified as ID if,
 - a) Proxy sample is part of private data.
 - b) Euclidean distance is within the specified threshold.
 - Uploads only ID proxy predictions to the server.
4. Client(s) download the aggregated proxy predictions from server for knowledge distillation.

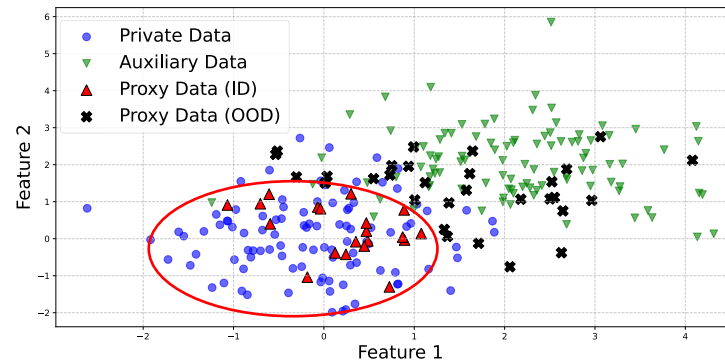
Repeat from step 1.

KMEANS-DRE VS. KULSIF-DRE



KMeans-DRE (Ours)

- Does not require auxiliary samples to learn private data distribution.
- Requires only training of centroids without large matrices and parameters generally required by statistical DREs.
- Smart two-stage client-side filtering that eliminates redundant computation in proxy data.

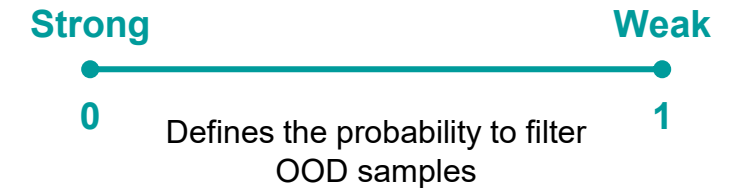


KuLSIF-DRE [1]

- Requires auxiliary samples generated synthetically; overloading clients with additional computation.
- Requires prior finetuning of performance-deciding factors, kernel width parameters, regularization settings, dataset extrema).
- Redundant operations due to the absence of two-staged client-side filter.

DREs Criteria for Classification

Depends on threshold to classify ID and OOD samples.



RESULTS

Bold represents the best results, and underline represents the second-best results. IndLearn refers to the average test accuracy when clients train independently.

INSIGHTS

- EdgeFD (Ours) consistently achieved **superior performance in all scenarios** (including best and second-best results).
- Under strong non-IID conditions, baseline methods exhibit significant performance limitations.
- For weak non-IID scenarios, baseline methods show improved performance compared to strong non-IID cases.
- Under IID conditions with uniform data distribution across clients, most methods achieve relatively high accuracy.

*“The performance improvement is attributed to the **KMeans-DRE** and **two-staged client-side filtering strategy of EdgeFD**, which effectively manages extreme data heterogeneity and filters OOD samples.”*

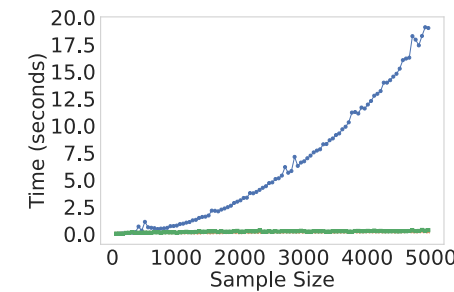
Strong Non-IID	MNIST	FashionMNIST	CIFAR10
IndLearn	10.00	10.00	10.00
FedMD [13]	88.71	64.63	15.78
FedED [15]	11.92	12.52	12.04
DS-FL [2]	35.25	35.98	12.07
FKD [16]	10.00	10.00	10.00
PLS [17]	10.00	10.00	10.00
Selective-FD [3]	94.68	75.31	80.98
EdgeFD (Ours)	98.92	88.55	82.57
Weak Non-IID	MNIST	FashionMNIST	CIFAR10
IndLearn	19.96	19.82	19.52
FedMD [13]	95.16	74.83	84.31
FedED [15]	60.26	37.12	56.13
DS-FL [2]	47.87	39.22	52.51
FKD [16]	19.98	19.71	19.51
PLS [17]	19.97	19.70	19.52
Selective-FD [3]	96.30	77.27	85.38
EdgeFD (Ours)	98.88	88.74	<u>84.88</u>
IID	MNIST	FashionMNIST	CIFAR10
IndLearn	98.83	88.80	85.53
FedMD [13]	<u>98.63</u>	<u>87.25</u>	86.31
FedED [15]	98.26	86.88	86.87
DS-FL [2]	98.56	86.62	85.82
FKD [16]	98.44	86.14	84.10
PLS [17]	98.48	86.52	84.77
Selective-FD [3]	98.60	87.16	86.06
EdgeFD (Ours)	99.08	89.90	<u>86.37</u>

Performance comparison of the proposed EdgeFD framework with the SoTA methods on various ML datasets under different non-IID conditions.

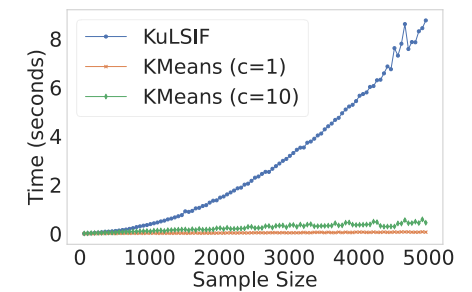
RESOURCE CONSUMPTION BY DENSITY RATIO ESTIMATORS

INSIGHTS

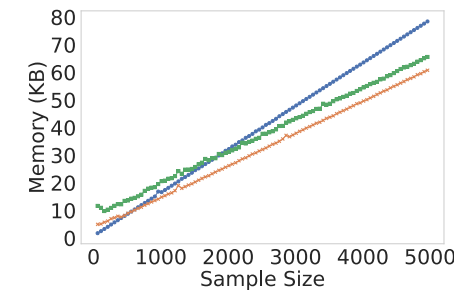
- KuLSIF-DRE exhibits **exponential** time complexity growth with increasing sample size for both phases.
- KMeans-DRE demonstrates **linear** time complexity with minimal slope across both phases.
- Memory complexity grows **linearly** with sample size of both DREs.
- Our KMeans-DRE shows **reduced memory consumption at large sample sizes**, highlighting its efficiency.



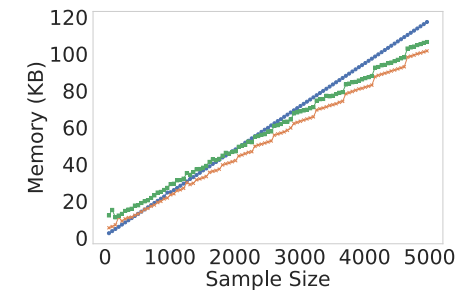
Learning Time



Estimation Time

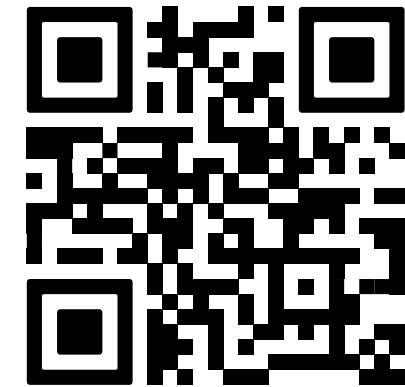
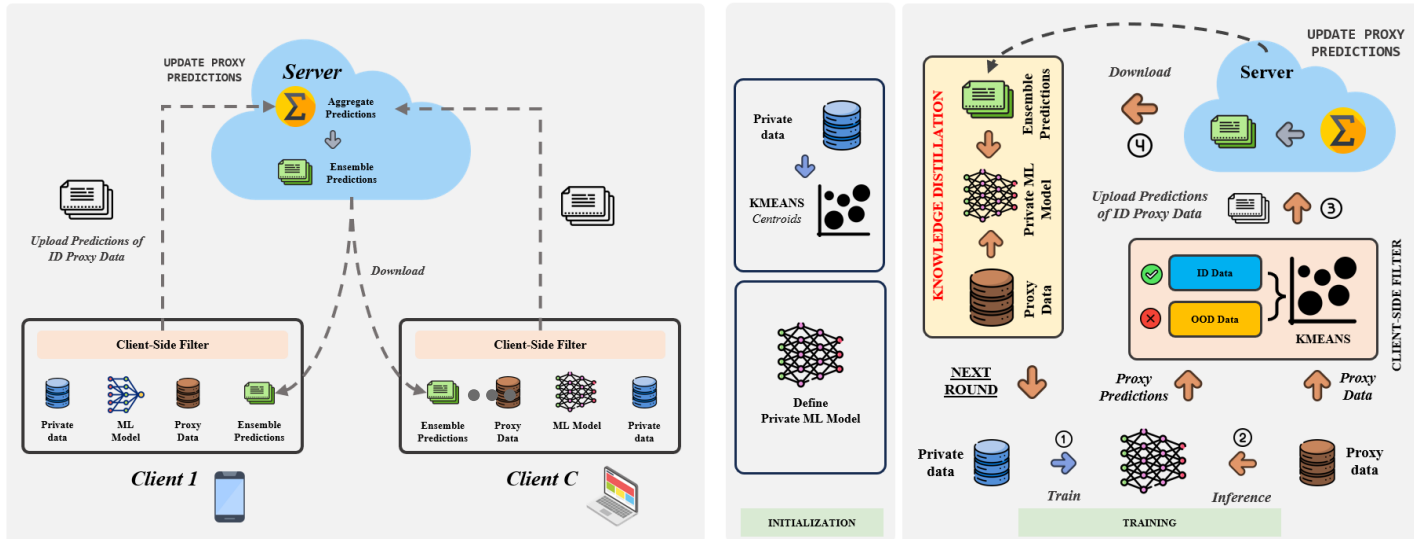


Learning Memory



Estimation Memory

Learning refers to learn the private data distribution.
Estimation refers to estimate the proxy data as ID or OOD.



[Open-Source Code](#)

- Addresses the challenges of **data heterogeneity** and **resource constraints** in edge-based federated environments.
- **Avoids** resource-intensive statistical DRE methods [1] and server-side selectors [2].
- **Minimizes** computational and memory overhead, through efficient **KMeans-based client-side filtering strategy**.
- **Requires** a small percentage (**20%**) of client's private data for effective collaborative learning.
- Demonstrate its **superior performance over state-of-the-art** federated distillation methods across heterogeneous data distribution.



THANK YOU FOR LISTENING 😊

QUESTIONS?

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16 Oct, 2025



SUPPLEMENTARY MATERIAL

RESOURCE CONSUMPTION BY DENSITY RATIO ESTIMATORS

INSIGHTS

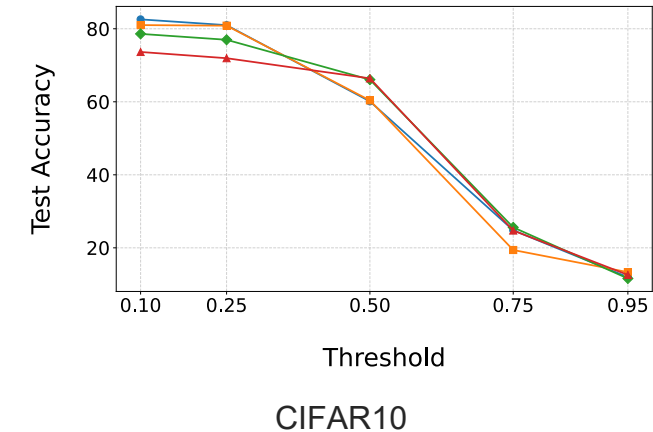
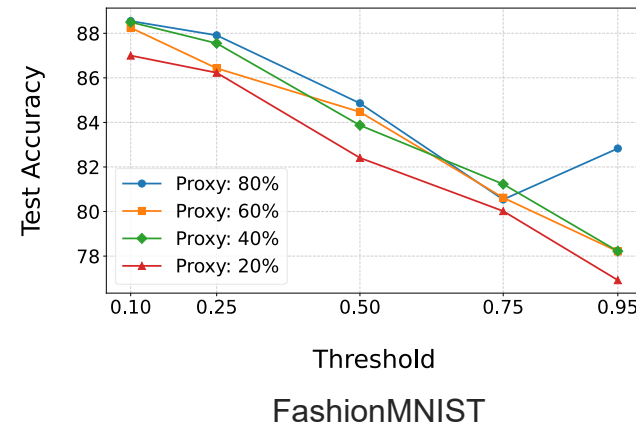
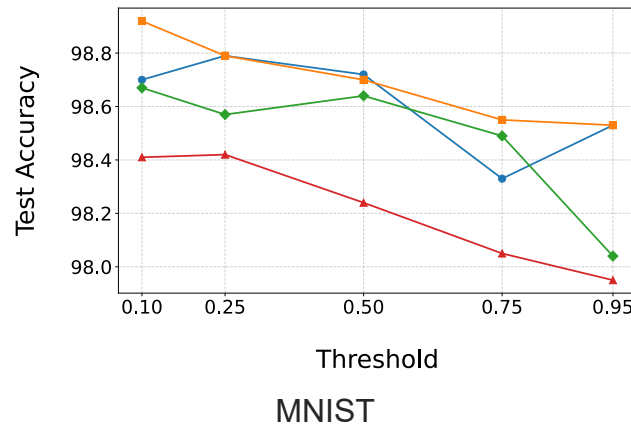
- The key parameters are auxiliary samples (m), private samples (n), test samples (t), and dimensionality of the sample data (d).
- KuLSIF-DRE exhibits **exponential** time complexity growth due to α -vector that requires K_{11} matrix inversion costing $O(m^3)$ time.
- After matrix inversion, to store α -vector and K_{11} , we need space for $m \times m$ entries $\rightarrow O(m^2)$.
- We **don't need to store all intermediate cubic operations** (they're computed and discarded on the fly).

TABLE IV: KuLSIF-DRE versus KMeans-DRE time and space complexity comparison.

<i>KuLSIF-DRE</i>	<i>Time</i>	<i>Space</i>
K_{11}	$\mathcal{O}(m^2 \cdot d)$	$\mathcal{O}(m^2)$
K_{12}	$\mathcal{O}(n \cdot m \cdot d)$	$\mathcal{O}(n \cdot m)$
α -vector	$\mathcal{O}(m^3)$	$\mathcal{O}(m^2)$
Learning	$\mathcal{O}(m^3 + m^2 \cdot d + n \cdot m \cdot d)$	$\mathcal{O}(m^2 + n \cdot m)$
Estimation	$\mathcal{O}(t \cdot (n + m) \cdot d)$	$\mathcal{O}(t \cdot (n + m))$
<i>KMeans-DRE</i>	<i>Time</i>	<i>Space</i>
Learning	$\mathcal{O}(k \cdot n \cdot c \cdot d)$	$\mathcal{O}(c \cdot d + n)$
Estimation	$\mathcal{O}(t \cdot c \cdot d)$	$\mathcal{O}(c \cdot d + t)$

Learning refers to learn the private data distribution.
Estimation refers to estimate the proxy data as ID or OOD.

FACTORS AFFECTING PERFORMANCE



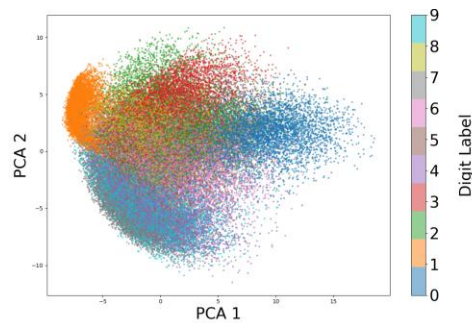
INSIGHTS — Proxy Data Percentage

- Increasing proxy data from 20% to 80% yields **minimal performance** gains.
- EdgeFD achieves effective knowledge distillation with just **20% of private data**, matching the performance of 80% proxy scenarios.

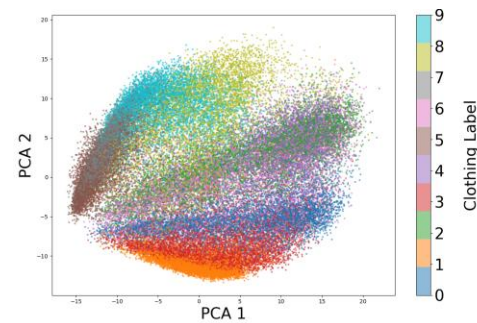
INSIGHTS — Threshold

- Higher thresholds correlate with decreased test accuracy; **increases** the OOD sample inclusion in ID proxy data.
- Higher threshold significantly **affects** complex dataset, e.g., CIFAR10, due to **inter-class feature overlap**.

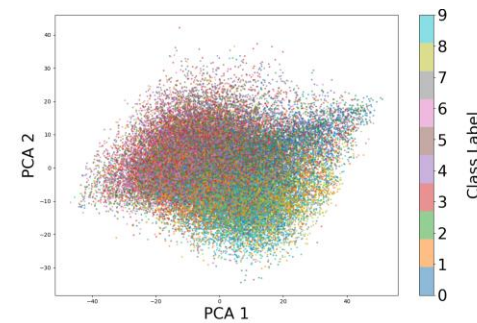
FACTORS AFFECTING PERFORMANCE



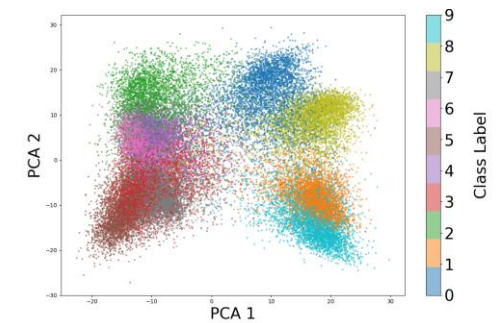
MNIST



FashionMNIST



CIFAR10

CIFAR10
(features extracted from pre-trained ResNet18)

INSIGHTS — Dataset Complexity

- MNIST and FashionMNIST contains **well-defined clusters** support effective learning and estimation of KMeans-DRE.
- Complex datasets like CIFAR10 containing three channel information and **overlapping high-dimensional** class distributions, limit DREs performance.

INSIGHTS — Feature Extraction

- Using pre-trained models (ResNet18 on ImageNet) to extract **low-dimensional features** from CIFAR-10 enables DREs to differentiate ID and OOD samples effectively.

HETEROGENEOUS MODELS

TABLE I: Customized ML architectures deployed across EdgeFD clients for MNIST and FashionMNIST datasets.

Client 1	Client 2	Client 3	Client 4	Client 5	Client 6	Client 7	Client 8	Client 9	Client 10
Conv2d(1, 10, 5)	Conv2d(1, 16, 3)	Conv2d(1, 10, 5)	Conv2d(1, 12, 3)	Conv2d(1, 8, 5)	Conv2d(1, 6, 7)	Conv2d(1, 32, 3)	Conv2d(1, 20, 5)	Conv2d(1, 8, 5)	Conv2d(1, 16, 3)
Conv2d(10, 20, 5)	Conv2d(16, 32, 3)	Conv2d(10, 20, 5)	Conv2d(12, 24, 3)	Conv2d(8, 16, 5)	Conv2d(6, 12, 5)	Conv2d(32, 64, 3)	Conv2d(20, 30, 5)	Conv2d(8, 16, 5)	Conv2d(16, 32, 3)
Linear(320, 50)	Conv2d(32, 64, 3)	Linear(320, 50)	Conv2d(24, 48, 3)	Linear(256, 100)	Linear(108, 50)	Linear(50176, 50)	Linear(480, 50)	Linear(256, 64)	Conv2d(32, 64, 3)
Linear(50, 10)	Linear(576, 50)	Linear(50, 10)	Linear(432, 100)	Linear(100, 50)	Linear(50, 10)	Linear(50, 10)	Linear(50, 10)	Linear(64, 32)	Linear(64, 100)
	Linear(50, 10)		Linear(100, 50)	Linear(50, 10)				Linear(32, 10)	Linear(100, 10)
			Linear(50, 10)						

TABLE II: Customized ML architectures deployed across clients in EdgeFD for the CIFAR-10 dataset.

Client 1	Client 2	Client 3	Client 4	Client 5	Client 6	Client 7	Client 8	Client 9	Client 10
$\begin{bmatrix} \text{Conv2d}(\cdot, 64, 3) \\ \text{BatchNorm2d}(64) \end{bmatrix} \times 2$	Conv2d(3, 64, 3) BatchNorm2d(64)								
$\begin{bmatrix} \text{Conv2d}(\cdot, 128, 3) \\ \text{BatchNorm2d}(128) \end{bmatrix} \times 2$	Conv2d(64, 128, 3) BatchNorm2d(128)	Conv2d(3, 64, 5) BatchNorm2d(64)	Conv2d(3, 64, 5) BatchNorm2d(64)	Conv2d(3, 32, 3) BatchNorm2d(32)	Conv2d(3, 32, 3) BatchNorm2d(32)	Conv2d(3, 32, 5) BatchNorm2d(32)	Conv2d(3, 32, 3) BatchNorm2d(32)	Conv2d(3, 64, 3) BatchNorm2d(64)	Conv2d(3, 64, 3) BatchNorm2d(64)
$\begin{bmatrix} \text{Conv2d}(\cdot, 256, 3) \\ \text{BatchNorm2d}(256) \end{bmatrix} \times 3$	Conv2d(128, 128, 3) BatchNorm2d(128)	Conv2d(64, 128, 5) BatchNorm2d(128)	Conv2d(64, 128, 5) BatchNorm2d(128)	Conv2d(32, 64, 3) BatchNorm2d(64)	Conv2d(32, 64, 3) BatchNorm2d(64)	Conv2d(32, 64, 5) BatchNorm2d(64)	Conv2d(32, 64, 3) BatchNorm2d(64)	Conv2d(64, 128, 3) BatchNorm2d(128)	Conv2d(64, 128, 3) BatchNorm2d(128)
Conv2d(128, 256, 3) BatchNorm2d(256)	Conv2d(128, 256, 3) BatchNorm2d(256)	Conv2d(128, 256, 3) BatchNorm2d(256)	Conv2d(128, 256, 3) BatchNorm2d(256)	Conv2d(64, 128, 3) BatchNorm2d(128)	Conv2d(64, 128, 3) BatchNorm2d(128)	Conv2d(64, 128, 3) BatchNorm2d(128)	Conv2d(64, 128, 3) BatchNorm2d(128)	Conv2d(128, 128, 3) BatchNorm2d(128)	Conv2d(128, 128, 3) BatchNorm2d(128)
Conv2d(128, 256, 1)	Conv2d(128, 256, 1)	Conv2d(256, 512, 3) BatchNorm2d(512)	Conv2d(256, 512, 3) BatchNorm2d(512)	Conv2d(128, 128, 3) BatchNorm2d(128)	Conv2d(128, 128, 3) BatchNorm2d(128)	Conv2d(128, 256, 3) BatchNorm2d(256)	Conv2d(128, 256, 3) BatchNorm2d(256)	Conv2d(64, 128, 1) Linear(8192, 512)	Conv2d(128, 256, 3) BatchNorm2d(128)
$\begin{bmatrix} \text{Conv2d}(\cdot, 512, 3) \\ \text{BatchNorm2d}(512) \end{bmatrix} \times 3$	Conv2d(256, 512, 3) BatchNorm2d(512)	Linear(16384, 1024) Linear(1024, 512)	Conv2d(256, 512, 3) BatchNorm2d(512)	Linear(32768, 512) Linear(512, 10)	Linear(8192, 512) Linear(512, 10)	Linear(16384, 1024) Linear(1024, 256)	Linear(8192, 512) Linear(512, 10)	Linear(512, 256) Linear(256, 10)	Linear(16384, 1024) Linear(1024, 10)
Linear(2048, 1024)	Linear(1024, 512)		Linear(512, 10)			Linear(4096, 512) Linear(512, 10)	Linear(256, 10)		
Linear(1024, 512)	Linear(512, 10)								
Linear(512, 10)									